EHSN-IOT

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Energy Harvesting Wireless Sensor Node

Internet of Things / Hazardous-Area Industrial Sensing | Rev A2 - Architecture, Atomic Requirements, Interfaces & Verification Baseline

Mission profile. Maintenance-free remote pipe sensor powered by ambient thermal and/or vibrational energy for oil-and-gas and manufacturing installations. The node operates on microwatt budgets, supports emergency alarm transmission from redundant storage, and is intended for Zone 1 hazardous-area deployment using an intrinsic-safety concept.

Engineering baseline. The hardware baseline combines piezoelectric and thermoelectric harvesting with ultra-low-power ARM Cortex-M0 processing, energy storage and a low-power industrial radio. Software is event-driven bare metal. LoRaWAN or WirelessHART variants use AES-128 payload protection. The stated ambient envelope is -40 C to +85 C, and EOL verifies harvester transducer efficiency.

Release boundary. Unspecified electrical limits, timing thresholds, environmental categories, certification interpretations and manufacturing acceptance values remain controlled TBC items until approved by the responsible engineering owner. Product requirements in this PRD define behavior, evidence and ownership without inventing absent numeric limits.

Definition	Controlled product interpretation
Product class	Energy Harvesting Wireless Sensor Node
Safety / assurance	Redundant energy storage supports the emergency alarm path. Intrinsic-safety energy limitation and Zone 1 installation evidence are release obligations; final certification details remain program controlled.
Lifecycle mission	15-year maintenance-free lifecycle intent with supercapacitor degradation modeling over thousands of charge cycles, energy-availability trending and field health evidence.
Primary architecture	Piezo/thermoelectric harvester + rectification/energy management + redundant storage + Cortex-M0 sensing + event-driven scheduling + LoRaWAN/WirelessHART radio + intrinsically safe enclosure.
Release philosophy	Energy, sensing, radio and hazardous-area claims are tied to the exact installation and hardware variant. Missing qualification or intrinsic-safety evidence blocks release of that variant; runtime low-energy behavior is handled separately by the energy state machine.

PRODUCT GOVERNANCE, ENERGY ACCOUNTING & REQUIREMENT QUALITY

A maintenance-free sensor can only claim availability when energy inflow, storage state, sensing demand and radio demand are all accounted for under the declared installation. Energy assumptions are therefore treated as controlled engineering inputs. The engineering objective for governance of harvested-energy assumptions, alarm reserve and release evidence is explicit ownership: integration teams must be able to identify the trusted inputs, permitted outputs and conditions that invalidate the result.

The dominant concern is an optimistic energy assumption creating silent loss of monitoring or emergency alarm capability. The verification package for this page therefore uses energy-budget audit; installation-envelope review; DVP traceability and records the tested revision/configuration so the result can be traced when the design changes.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-GOV-001	governance of harvested-energy assumptions, alarm reserve and release evidence shall be implemented with explicit ownership, validity rules and bounded behavior for the released configuration.	Energy-budget audit.
EHSN-GOV-002	The design shall detect or contain an optimistic energy assumption creating silent loss of monitoring or emergency alarm capability before it can violate the stated trust, availability or interface boundary.	Targeted governance of harvested-energy assumptions, alarm reserve and release evidence fault injection and diagnostic-state correlation.
EHSN-GOV-003	Release of this function shall require energy-budget audit; installation-envelope review; DVP traceability; an unresolved result keeps the affected configuration open.	governance of harvested-energy assumptions, alarm reserve and release evidence release checklist and configuration-status audit.

Verification, margin and future-change rule

For governance of harvested-energy assumptions, alarm reserve and release evidence, reassess the release argument when any of these change: harvester profile, sensor duty, radio profile, storage technology, firmware schedule or installation. Regression for governance of harvested-energy assumptions, alarm reserve and release evidence shall target only the assumptions changed and shall preserve traceability to the prior energy-budget audit evidence rather than repeating unrelated qualification by default.

DEPLOYMENT BOUNDARY, Zone 1 CONTEXT & EXPLICIT EXCLUSIONS

the boundary between the sensor node, pipe asset, hazardous-area installation and network is a product-owned mechanism with installation dependencies documented separately. The node observes remote pipe assets in oil-and-gas and manufacturing facilities. Intrinsic-safety and radio-installation responsibilities are separated from the host process so the node does not silently become a process-control actuator. For the boundary between the sensor node, pipe asset, hazardous-area installation and network, that separation keeps deployment choices visible without turning them into hidden assumptions inside the product claim.

Risk analysis centers on unclear installation responsibility or an external accessory defeating the intrinsic-safety energy boundary. Instead of inventing a missing margin, the program keeps the unresolved parameter visible and closes it through interface review; hazardous-area boundary audit; misuse-case test; operational containment and product-release approval remain separate decisions.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-BND-001	The product shall preserve the intended function of the boundary between the sensor node, pipe asset, hazardous-area installation and network across the operational states defined for the installation.	Interface review.
EHSN-BND-002	A fault involving unclear installation responsibility or an external accessory defeating the intrinsic-safety energy boundary shall produce a diagnosable state and shall not silently expand authority beyond the approved boundary.	Negative-path test for the boundary between the sensor node, pipe asset, hazardous-area installation and network with recovery-state observation.
EHSN-BND-003	Qualification shall close interface review; hazardous-area boundary audit; misuse-case test with results traceable to the tested hardware, software and configuration.	Variant evidence review for the boundary between the sensor node, pipe asset, hazardous-area installation and network and production-approval gate.

Verification, margin and future-change rule

For the boundary between the sensor node, pipe asset, hazardous-area installation and network, reassess the release argument when any of these change: mounting, sensor interface, antenna, enclosure entry, network gateway or site classification. Regression for the boundary between the sensor node, pipe asset, hazardous-area installation and network shall target only the assumptions changed and shall preserve traceability to the prior interface review evidence rather than repeating unrelated qualification by default.

OPERATING MODES, COLD START, MEASUREMENT & ALARM RESERVE

The energy state machine sequences accumulation, cold start, sleep, measurement, local processing, radio exchange and emergency alarm according to measured stored-energy state. Entry into a higher-energy activity requires the corresponding energy qualification to be valid, and interrupted cycles return to a defined low-energy state rather than repeatedly consuming the reserve.

Failure analysis addresses entering a high-energy mode without enough reserve to complete the operation or preserve an alarm opportunity. Diagnostic outputs must distinguish bad input, internal malfunction and incomplete engineering evidence, while state-machine energy test; interrupted-cycle test; reserve-threshold review provides the objective basis for closing the mechanism.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-MOD-001	The energy state machine shall gate accumulation, cold start, sleep, measurement, processing, radio activity and emergency alarm states using qualified stored-energy conditions.	State-machine energy test.
EHSN-MOD-002	A transition into a higher-energy activity shall be prevented when the available reserve is insufficient to complete the operation and preserve the protected alarm opportunity.	Boundary-condition campaign for the operating-mode energy state machine with event-log review.
EHSN-MOD-003	Energy-state-machine changes shall require state-transition, interrupted-cycle and reserve-threshold regression before production equivalence.	the operating-mode energy state machine traceability audit with regression-closure review.

Verification, margin and future-change rule

For the operating-mode energy state machine, reassess the release argument when any of these change: harvester power, storage capacitance, firmware mode thresholds, sensor load or radio profile. Regression for the operating-mode energy state machine shall target only the assumptions changed and shall preserve traceability to the prior state-machine energy test evidence rather than repeating unrelated qualification by default.

ENERGY-HARVESTING NODE ARCHITECTURE

Across startup, normal service, degraded operation and authorized recovery, the end-to-end harvester, storage, sensing, compute and radio architecture must preserve the behavior described here. The released architecture links piezoelectric and/or thermoelectric energy capture to conditioning, storage, an ultra-low-power Cortex-M0, sensors and a low-power industrial radio. Each domain exposes its energy demand or contribution to the system budget. State transitions for the end-to-end harvester, storage, sensing, compute and radio architecture are verified from observable conditions rather than inferred from nominal success alone.

The assurance case shall demonstrate that parasitic leakage, hidden loads or sequencing dependencies cannot consume enough harvested energy to invalidate the released energy budget. Verification uses block-diagram review, quiescent-current measurement and end-to-end source-replay testing across nominal, weak-source and injected-fault conditions.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-ARC-001	The architecture shall provide observability for the end-to-end harvester, storage, sensing, compute and radio architecture sufficient to distinguish valid operation from degraded or compromised conditions.	Block-diagram review.
EHSN-ARC-002	When an undocumented parasitic path or sequencing dependency consuming the available harvested energy occurs, containment shall follow the defined policy and retain the evidence needed for diagnosis.	Injected-failure test for the end-to-end harvester, storage, sensing, compute and radio architecture with containment and restart evidence.
EHSN-ARC-003	Acceptance shall be based on block-diagram review; quiescent-current measurement; end-to-end energy test, including configuration traceability where it affects the conclusion.	the end-to-end harvester, storage, sensing, compute and radio architecture configuration baseline and qualification-status review.

Verification, margin and future-change rule

For the end-to-end harvester, storage, sensing, compute and radio architecture, reassess the release argument when any of these change: harvester variant, PMIC, storage, MCU, sensor, radio, PCB leakage or connectorization. Regression for the end-to-end harvester, storage, sensing, compute and radio architecture shall target only the assumptions changed and shall preserve traceability to the prior block-diagram review evidence rather than repeating unrelated qualification by default.

PIEZOELECTRIC HARVESTER INPUT & MECHANICAL COUPLING

Engineering ownership of piezoelectric energy conversion includes its dependencies and failure evidence. Vibration harvesting depends on mechanical coupling, frequency content and transducer loading. The product shall characterize energy capture over the approved mounting condition rather than quote an isolated bench peak. The release record therefore ties piezoelectric energy conversion to the exact implementation and configured external services used in verification.

The release risk is resonance shift, poor mounting or fatigue reducing harvested energy below the monitoring budget. Evidence from shaker test; mount-variation study; efficiency map is reviewed together with interface state and recovery behavior; an engineering gap blocks the affected variant rather than being disguised as a runtime derating strategy.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-PZO-001	piezoelectric energy conversion shall be dimensioned and configured for the declared mission profile without relying on undocumented external behavior.	Shaker test.
EHSN-PZO-002	The consequence of resonance shift, poor mounting or fatigue reducing harvested energy below the monitoring budget shall be bounded so that recovery, fail-closed behavior or service action remains deterministic.	Misuse-case test for piezoelectric energy conversion plus diagnostic and recovery verification.
EHSN-PZO-003	Engineering release shall close shaker test; mount-variation study; efficiency map; supplier, layout, firmware or installation changes that alter the argument require focused retest.	Production-equivalence decision record for piezoelectric energy conversion.

Verification, margin and future-change rule

For piezoelectric energy conversion, reassess the release argument when any of these change: piezo element, clamp, proof mass, mounting torque, pipe vibration spectrum or rectifier loading. Regression for piezoelectric energy conversion shall target only the assumptions changed and shall preserve traceability to the prior shaker test evidence rather than repeating unrelated qualification by default.

THERMOELECTRIC HARVESTER & THERMAL-GRADIENT COUPLING

The product treats thermoelectric energy conversion as a controlled capability rather than a generic platform feature. Thermoelectric harvesting depends on sustained temperature gradient and thermal resistance on both sides of the generator. Ambient temperature alone is not a sufficient availability predictor. Any dependency of thermoelectric energy conversion is named in the interface or configuration baseline so its loss can be distinguished from an internal defect.

Thermoelectric-harvester validation addresses reduced thermal gradient, fouling and interface aging that can collapse usable harvested power. Thermal-gradient testing, interface-aging exposure and power-versus-delta-T mapping quantify the available margin.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-TEG-001	The released implementation of thermoelectric energy conversion shall expose the status required by diagnostics and lifecycle management without disclosing protected secrets or unsafe controls.	Thermal-gradient test.
EHSN-TEG-002	Reduced thermal gradient, fouling or interface aging that collapses usable harvested power shall be reflected in the available-energy state and shall not be masked as nominal harvesting.	Fault stimulation of thermoelectric energy conversion with interface-state and audit-record review.
EHSN-TEG-003	Objective evidence for this requirement shall include thermal-gradient test; interface aging; power-versus-delta-T map, using pass/fail criteria owned by the product verification plan.	DVP closure for thermoelectric energy conversion and controlled release-board review.

Verification, margin and future-change rule

For thermoelectric energy conversion, reassess the release argument when any of these change: TEG, thermal interface, heat spreader, pipe temperature, ambient flow or enclosure conduction. Regression for thermoelectric energy conversion shall target only the assumptions changed and shall preserve traceability to the prior thermal-gradient test evidence rather than repeating unrelated qualification by default.

RECTIFICATION, INPUT PROTECTION & LEAKAGE CONTROL

Low harvested power makes forward drop, reverse leakage and clamp leakage first-order design parameters. Protection must survive expected transients without consuming the energy that the node depends on in normal operation. The engineering objective for harvester rectification and input protection is explicit ownership: integration teams must be able to identify the trusted inputs, permitted outputs and conditions that invalidate the result.

The dominant concern is rectifier or protection leakage preventing cold start or overstress damaging the harvester interface. The verification package for this page therefore uses I-V characterization; leakage test; transient stress test and records the tested revision/configuration so the result can be traced when the design changes.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-REC-001	harvester rectification and input protection shall be implemented with explicit ownership, validity rules and bounded behavior for the released configuration.	I-v characterization.
EHSN-REC-002	The design shall detect or contain rectifier or protection leakage preventing cold start or overstress damaging the harvester interface before it can violate the stated trust, availability or interface boundary.	Abnormal-condition test for harvester rectification and input protection and degraded-state trace review.
EHSN-REC-003	Release of this function shall require I-V characterization; leakage test; transient stress test; an unresolved result keeps the affected configuration open.	harvester rectification and input protection release checklist and configuration-status audit.

Verification, margin and future-change rule

For harvester rectification and input protection, reassess the release argument when any of these change: rectifier topology, protection diode, input capacitor, PCB contamination, temperature or harvester impedance. Regression for harvester rectification and input protection shall target only the assumptions changed and shall preserve traceability to the prior I-V characterization evidence rather than repeating unrelated qualification by default.

ENERGY-MANAGEMENT / MPPT OPERATING POINT CONTROL

energy-management and source loading is a product-owned mechanism with installation dependencies documented separately. The energy-management stage presents a load that allows the active harvester to deliver useful energy across changing vibration or thermal conditions. Exact MPPT method and thresholds remain controlled design parameters. For energy-management and source loading, that separation keeps deployment choices visible without turning them into hidden assumptions inside the product claim.

Risk analysis centers on control hunting, excessive quiescent current or wrong loading reducing net harvested energy. Instead of inventing a missing margin, the program keeps the unresolved parameter visible and closes it through source-emulator test; efficiency map; quiescent-current measurement; operational containment and product-release approval remain separate decisions.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-MPPT-001	The product shall preserve the intended function of energy-management and source loading across the operational states defined for the installation.	Source-emulator test.
EHSN-MPPT-002	A fault involving control hunting, excessive quiescent current or wrong loading reducing net harvested energy shall produce a diagnosable state and shall not silently expand authority beyond the approved boundary.	Stress/fault test for energy-management and source loading with recovery and non-bypass evidence.
EHSN-MPPT-003	Qualification shall close source-emulator test; efficiency map; quiescent-current measurement with results traceable to the tested hardware, software and configuration.	Variant evidence review for energy-management and source loading and production-approval gate.

Verification, margin and future-change rule

For energy-management and source loading, reassess the release argument when any of these change: PMIC, firmware policy, input capacitor, set point, harvester impedance or temperature. Regression for energy-management and source loading shall target only the assumptions changed and shall preserve traceability to the prior source-emulator test evidence rather than repeating unrelated qualification by default.

REDUNDANT ENERGY STORAGE CAPACITORS & ENERGY PATH

Redundant energy-storage capacitors support the emergency alarm path even when one storage element degrades or disconnects. Isolation prevents a failed element from discharging the healthy reserve, and the energy manager receives explicit storage-health information so degraded redundancy changes scheduling and diagnostics without masking the remaining reserve.

Failure analysis addresses single capacitor short/open or leakage fault removing the entire alarm-energy reserve. Diagnostic outputs must distinguish bad input, internal malfunction and incomplete engineering evidence, while single-fault injection; charge/discharge test; reserve-energy measurement provides the objective basis for closing the mechanism.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-STR-001	Redundant storage shall isolate single-element faults so one failed capacitor cannot discharge or disable the entire protected alarm-energy reserve.	Single-fault injection.
EHSN-STR-002	A capacitor short, open circuit or excessive leakage shall be detected or bounded so the remaining healthy storage path retains the energy allocated to the alarm function.	Targeted redundant energy storage and emergency reserve fault injection and diagnostic-state correlation.
EHSN-STR-003	Storage-topology or capacitor changes shall require single-fault isolation, charge/discharge and reserve-energy validation before production equivalence.	redundant energy storage and emergency reserve traceability audit with regression-closure review.

Verification, margin and future-change rule

For redundant energy storage and emergency reserve, reassess the release argument when any of these change: capacitor technology, isolation element, ESR, leakage, layout or charge control. Regression for redundant energy storage and emergency reserve shall target only the assumptions changed and shall preserve traceability to the prior single-fault injection evidence rather than repeating unrelated qualification by default.

SUPERCAPACITOR STORAGE, ESR & LEAKAGE AGING

Across startup, normal service, degraded operation and authorized recovery, supercapacitor lifecycle behavior must preserve the behavior described here. The lifecycle baseline includes supercapacitor degradation over thousands of charge cycles. Capacity, ESR and leakage are therefore tracked as energy-availability variables rather than only component health statistics. State transitions for supercapacitor lifecycle behavior are verified from observable conditions rather than inferred from nominal success alone.

The storage-aging assurance case shall demonstrate that capacity fade, ESR growth and leakage cannot cause the energy estimator to overstate deliverable alarm energy. Verification uses cycle-aging, pulse-delivery and ESR/leakage correlation testing across temperature and state-of-charge conditions.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-SCP-001	The architecture shall provide observability for supercapacitor lifecycle behavior sufficient to distinguish valid operation from degraded or compromised conditions.	Cycle-aging test.
EHSN-SCP-002	When aging causing apparent state of charge to overestimate deliverable alarm energy occurs, containment shall follow the defined policy and retain the evidence needed for diagnosis.	Negative-path test for supercapacitor lifecycle behavior with recovery-state observation.
EHSN-SCP-003	Acceptance shall be based on cycle-aging test; pulse-delivery test; ESR/leakage trend correlation, including configuration traceability where it affects the conclusion.	supercapacitor lifecycle behavior configuration baseline and qualification-status review.

Verification, margin and future-change rule

For supercapacitor lifecycle behavior, reassess the release argument when any of these change: supercapacitor family, voltage derating, temperature, charge window, balancing or replacement model. Regression for supercapacitor lifecycle behavior shall target only the assumptions changed and shall preserve traceability to the prior cycle-aging test evidence rather than repeating unrelated qualification by default.

REGULATION, BROWNOUT & POWER-DOMAIN SEQUENCING

Engineering ownership of regulated rails and brownout behavior includes its dependencies and failure evidence. MCU, sensor and radio rails start only when stored energy can support the intended transaction. Brownout resets must leave storage accounting and event state consistent. The release record therefore ties regulated rails and brownout behavior to the exact implementation and configured external services used in verification.

The release risk is oscillatory restart consuming all stored energy or corrupting event/energy state. Evidence from brownout sweep; repeated-start test; NVM consistency review is reviewed together with interface state and recovery behavior; an engineering gap blocks the affected variant rather than being disguised as a runtime derating strategy.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-REG-001	regulated rails and brownout behavior shall be dimensioned and configured for the declared mission profile without relying on undocumented external behavior.	Brownout sweep.
EHSN-REG-002	The consequence of oscillatory restart consuming all stored energy or corrupting event/energy state shall be bounded so that recovery, fail-closed behavior or service action remains deterministic.	Boundary-condition campaign for regulated rails and brownout behavior with event-log review.
EHSN-REG-003	Engineering release shall close brownout sweep; repeated-start test; NVM consistency review; supplier, layout, firmware or installation changes that alter the argument require focused retest.	Production-equivalence decision record for regulated rails and brownout behavior.

Verification, margin and future-change rule

For regulated rails and brownout behavior, reassess the release argument when any of these change: regulator, threshold, hysteresis, rail sequencing, hold-up or firmware startup. Regression for regulated rails and brownout behavior shall target only the assumptions changed and shall preserve traceability to the prior brownout sweep evidence rather than repeating unrelated qualification by default.

MICROWATT POWER BUDGET & QUIESCENT-CURRENT OWNERSHIP

Operation is based on a microwatt-level system energy budget. Sleep leakage, sensing, wake processing, radio bursts and self-discharge are assigned to named budget owners with controlled measurement conditions. The budget identifies external energy-source assumptions separately from internally measured consumption so weak-source behavior can be verified without inventing unsupported margins.

Energy-budget verification addresses small unowned quiescent loads that can accumulate into a mission-level deficit. Current profiling, budget reconciliation and temperature-corner measurements shall close every material load contribution.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-BUD-001	The released implementation of the microwatt-level energy budget shall expose the status required by diagnostics and lifecycle management without disclosing protected secrets or unsafe controls.	Current profiling.
EHSN-BUD-002	Unowned quiescent loads that accumulate into a mission-level energy deficit shall be identified by budget reconciliation and shall prevent release until the deficit is resolved.	Injected-failure test for the microwatt-level energy budget with containment and restart evidence.
EHSN-BUD-003	Objective evidence for this requirement shall include current profiling; budget reconciliation; temperature-corner measurement, using pass/fail criteria owned by the product verification plan.	DVP closure for the microwatt-level energy budget and controlled release-board review.

Verification, margin and future-change rule

For the microwatt-level energy budget, reassess the release argument when any of these change: MCU sleep mode, sensor, PMIC, radio, PCB leakage, pull resistors or debug circuitry. Regression for the microwatt-level energy budget shall target only the assumptions changed and shall preserve traceability to the prior current profiling evidence rather than repeating unrelated qualification by default.

COLD-START ENERGY ACCUMULATION & FIRST-BOOT POLICY

From a depleted state, the node accumulates enough energy to initialize power management, validate state and begin sensing without repeated partial boots. Radio transmission is deferred until its energy reserve is demonstrably available. The engineering objective for cold-start behavior from an empty energy store is explicit ownership: integration teams must be able to identify the trusted inputs, permitted outputs and conditions that invalidate the result.

The dominant concern is boot chatter wasting scarce energy and preventing the node from ever reaching a stable monitoring state. The verification package for this page therefore uses zero-energy startup test; weak-source test; boot-attempt counter review and records the tested revision/configuration so the result can be traced when the design changes.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-CST-001	cold-start behavior from an empty energy store shall be implemented with explicit ownership, validity rules and bounded behavior for the released configuration.	Zero-energy startup test.
EHSN-CST-002	The design shall detect or contain boot chatter wasting scarce energy and preventing the node from ever reaching a stable monitoring state before it can violate the stated trust, availability or interface boundary.	Misuse-case test for cold-start behavior from an empty energy store plus diagnostic and recovery verification.
EHSN-CST-003	Release of this function shall require zero-energy startup test; weak-source test; boot-attempt counter review; an unresolved result keeps the affected configuration open.	cold-start behavior from an empty energy store release checklist and configuration-status audit.

Verification, margin and future-change rule

For cold-start behavior from an empty energy store, reassess the release argument when any of these change: startup threshold, bootloader, NVM initialization, sensor warm-up or radio policy. Regression for cold-start behavior from an empty energy store shall target only the assumptions changed and shall preserve traceability to the prior zero-energy startup test evidence rather than repeating unrelated qualification by default.

DUTY CYCLING, WAKE SOURCES & ENERGY-AWARE SCHEDULING

duty cycling and energy-aware scheduling is a product-owned mechanism with installation dependencies documented separately. Event-driven bare-metal execution minimizes active time. Periodic maintenance tasks are scheduled only when their value and energy cost are justified by the mission and available reserve. For duty cycling and energy-aware scheduling, that separation keeps deployment choices visible without turning them into hidden assumptions inside the product claim.

Risk analysis centers on background tasks or excessive wakeups eroding the energy budget without improving monitoring. Instead of inventing a missing margin, the program keeps the unresolved parameter visible and closes it through wake-cause logging; energy-per-task profiling; long-duration source replay; operational containment and product-release approval remain separate decisions.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-DUT-001	The product shall preserve the intended function of duty cycling and energy-aware scheduling across the operational states defined for the installation.	Wake-cause logging.
EHSN-DUT-002	A fault involving background tasks or excessive wakeups eroding the energy budget without improving monitoring shall produce a diagnosable state and shall not silently expand authority beyond the approved boundary.	Fault stimulation of duty cycling and energy-aware scheduling with interface-state and audit-record review.
EHSN-DUT-003	Qualification shall close wake-cause logging; energy-per-task profiling; long-duration source replay with results traceable to the tested hardware, software and configuration.	Variant evidence review for duty cycling and energy-aware scheduling and production-approval gate.

Verification, margin and future-change rule

For duty cycling and energy-aware scheduling, reassess the release argument when any of these change: timer cadence, sensor interrupt, housekeeping task, radio check-in or calibration routine. Regression for duty cycling and energy-aware scheduling shall target only the assumptions changed and shall preserve traceability to the prior wake-cause logging evidence rather than repeating unrelated qualification by default.

EMERGENCY ALARM ENERGY RESERVE & TRANSMISSION ATTEMPT

A protected portion of stored energy is reserved for the emergency alarm transaction. The energy manager owns this reserve state, and nonessential sensing, logging or routine radio traffic is curtailed before the reserve is consumed. Alarm eligibility is based on deliverable pulse energy under the current storage condition, not only on open-circuit voltage.

Failure analysis addresses routine measurements or retries consuming the last energy needed for alarm transmission. Diagnostic outputs must distinguish bad input, internal malfunction and incomplete engineering evidence, while reserve-depletion test; alarm-at-low-energy test; retry-policy verification provides the objective basis for closing the mechanism.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-ALR-001	The emergency-alarm reserve shall be explicitly protected from routine sensing, logging and retry traffic once stored energy approaches the controlled reserve threshold.	Reserve-depletion test.
EHSN-ALR-002	Routine measurements or communication retries shall not consume the final energy required for the approved emergency alarm transaction.	Abnormal-condition test for the emergency alarm reserve and degraded-state trace review.
EHSN-ALR-003	Alarm-reserve policy changes shall require reserve-depletion, low-energy alarm and retry-policy verification before release.	the emergency alarm reserve traceability audit with regression-closure review.

Verification, margin and future-change rule

For the emergency alarm reserve, reassess the release argument when any of these change: alarm packet, radio, transmit power, retry count, storage capacity or reserve threshold. Regression for the emergency alarm reserve shall target only the assumptions changed and shall preserve traceability to the prior reserve-depletion test evidence rather than repeating unrelated qualification by default.

ENERGY-AVAILABILITY ESTIMATION & CONFIDENCE

Across startup, normal service, degraded operation and authorized recovery, energy-availability estimation must preserve the behavior described here. The node estimates whether enough energy exists for measurement, processing and communication using storage voltage plus technology-specific behavior such as ESR, leakage and temperature. State transitions for energy-availability estimation are verified from observable conditions rather than inferred from nominal success alone.

The energy-state estimator shall be validated so voltage-only behavior cannot overstate usable energy when storage is aged, cold or under pulse load. Verification uses estimator correlation, pulse testing and temperature/aging matrices.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-EAV-001	The architecture shall provide observability for energy-availability estimation sufficient to distinguish valid operation from degraded or compromised conditions.	Energy estimator correlation.
EHSN-EAV-002	When voltage-only estimation overstating usable energy under aged or cold storage conditions occurs, containment shall follow the defined policy and retain the evidence needed for diagnosis.	Stress/fault test for energy-availability estimation with recovery and non-bypass evidence.
EHSN-EAV-003	Acceptance shall be based on energy estimator correlation; pulse test; temperature/aging matrix, including configuration traceability where it affects the conclusion.	energy-availability estimation configuration baseline and qualification-status review.

Verification, margin and future-change rule

For energy-availability estimation, reassess the release argument when any of these change: storage model, voltage sensor, ESR estimate, temperature input or calibration. Regression for energy-availability estimation shall target only the assumptions changed and shall preserve traceability to the prior energy estimator correlation evidence rather than repeating unrelated qualification by default.

ULTRA-LOW-POWER ARM Cortex-M0 COMPUTE BASELINE

Engineering ownership of the arm Cortex-M0 compute platform includes its dependencies and failure evidence. The Cortex-M0 executes only the processing required for event detection, energy management and secure communication. Clocking, memory retention and peripheral use are optimized around the measured energy budget. The release record therefore ties the arm Cortex-M0 compute platform to the exact implementation and configured external services used in verification.

The release risk is firmware configuration leaving high-frequency clocks or peripherals active during nominal sleep. Evidence from power-state audit; clock/peripheral test; current profiling is reviewed together with interface state and recovery behavior; an engineering gap blocks the affected variant rather than being disguised as a runtime derating strategy.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-MCU-001	the ARM Cortex-M0 compute platform shall be dimensioned and configured for the declared mission profile without relying on undocumented external behavior.	Power-state audit.
EHSN-MCU-002	The consequence of firmware configuration leaving high-frequency clocks or peripherals active during nominal sleep shall be bounded so that recovery, fail-closed behavior or service action remains deterministic.	Targeted the arm Cortex-M0 compute platform fault injection and diagnostic-state correlation.
EHSN-MCU-003	Engineering release shall close power-state audit; clock/peripheral test; current profiling; supplier, layout, firmware or installation changes that alter the argument require focused retest.	Production-equivalence decision record for the arm Cortex-M0 compute platform.

Verification, margin and future-change rule

For the arm Cortex-M0 compute platform, reassess the release argument when any of these change: MCU variant, clock tree, retention RAM, debug configuration, compiler or peripheral mapping. Regression for the arm Cortex-M0 compute platform shall target only the assumptions changed and shall preserve traceability to the prior power-state audit evidence rather than repeating unrelated qualification by default.

EVENT-DRIVEN BARE-METAL SCHEDULER

The product treats bare-metal event scheduling as a controlled capability rather than a generic platform feature. The software baseline is event driven rather than a continuously running task framework. Interrupt work is bounded and deferred processing must return the system to the lowest valid power state. Any dependency of bare-metal event scheduling is named in the interface or configuration baseline so its loss can be distinguished from an internal defect.

Firmware timing verification addresses interrupt storms, re-entrant handlers and long processing loops that can prevent timely return to sleep. Event-storm testing, execution-time measurement and stack/latency review demonstrate bounded active time.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-SCH-001	The released implementation of bare-metal event scheduling shall expose the status required by diagnostics and lifecycle management without disclosing protected secrets or unsafe controls.	Event-storm test.
EHSN-SCH-002	Interrupt storms, re-entrant handlers or excessive processing time shall be bounded so firmware returns to the required low-power state within the verified energy budget.	Negative-path test for bare-metal event scheduling with recovery-state observation.
EHSN-SCH-003	Objective evidence for this requirement shall include event-storm test; execution-time measurement; stack/latency review, using pass/fail criteria owned by the product verification plan.	DVP closure for bare-metal event scheduling and controlled release-board review.

Verification, margin and future-change rule

For bare-metal event scheduling, reassess the release argument when any of these change: interrupt source, handler, event queue, timer, compiler optimization or firmware feature. Regression for bare-metal event scheduling shall target only the assumptions changed and shall preserve traceability to the prior event-storm test evidence rather than repeating unrelated qualification by default.

THRESHOLD TRIGGER PIPELINE & EVENT QUALIFICATION

Thresholds initiate higher-cost processing only after the sensing path has enough evidence to distinguish a meaningful event from noise. Debounce and persistence logic are controlled by the sensing application. The engineering objective for threshold-triggered measurement and alarm logic is explicit ownership: integration teams must be able to identify the trusted inputs, permitted outputs and conditions that invalidate the result.

The dominant concern is noise bursts repeatedly waking the node or real events being suppressed by stale threshold state. The verification package for this page therefore uses signal replay; noise susceptibility test; threshold-boundary verification and records the tested revision/configuration so the result can be traced when the design changes.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-TRG-001	threshold-triggered measurement and alarm logic shall be implemented with explicit ownership, validity rules and bounded behavior for the released configuration.	Signal replay.
EHSN-TRG-002	The design shall detect or contain noise bursts repeatedly waking the node or real events being suppressed by stale threshold state before it can violate the stated trust, availability or interface boundary.	Boundary-condition campaign for threshold-triggered measurement and alarm logic with event-log review.
EHSN-TRG-003	Release of this function shall require signal replay; noise susceptibility test; threshold-boundary verification; an unresolved result keeps the affected configuration open.	threshold-triggered measurement and alarm logic release checklist and configuration-status audit.

Verification, margin and future-change rule

For threshold-triggered measurement and alarm logic, reassess the release argument when any of these change: sensor, filter, threshold, debounce, persistence time or calibration. Regression for threshold-triggered measurement and alarm logic shall target only the assumptions changed and shall preserve traceability to the prior signal replay evidence rather than repeating unrelated qualification by default.

SENSOR FRONT-END, EXCITATION & LOW-POWER MEASUREMENT

the low-power sensor acquisition path is a product-owned mechanism with installation dependencies documented separately. Remote pipe monitoring may use application-specific sensing elements, but their excitation, warm-up, accuracy and energy cost must be explicit in the released variant. The PRD does not invent a sensor technology not supplied by the program. For the low-power sensor acquisition path, that separation keeps deployment choices visible without turning them into hidden assumptions inside the product claim.

Risk analysis centers on sensor warm-up or front-end leakage dominating the available energy or producing invalid event decisions. Instead of inventing a missing margin, the program keeps the unresolved parameter visible and closes it through sensor-current profile; accuracy/energy trade study; open/short fault test; operational containment and product-release approval remain separate decisions.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-SEN-001	The product shall preserve the intended function of the low-power sensor acquisition path across the operational states defined for the installation.	Sensor-current profile.
EHSN-SEN-002	A fault involving sensor warm-up or front-end leakage dominating the available energy or producing invalid event decisions shall produce a diagnosable state and shall not silently expand authority beyond the approved boundary.	Injected-failure test for the low-power sensor acquisition path with containment and restart evidence.
EHSN-SEN-003	Qualification shall close sensor-current profile; accuracy/energy trade study; open/short fault test with results traceable to the tested hardware, software and configuration.	Variant evidence review for the low-power sensor acquisition path and production-approval gate.

Verification, margin and future-change rule

For the low-power sensor acquisition path, reassess the release argument when any of these change: sensor technology, excitation, ADC, filter, sampling plan or calibration. Regression for the low-power sensor acquisition path shall target only the assumptions changed and shall preserve traceability to the prior sensor-current profile evidence rather than repeating unrelated qualification by default.

THRESHOLD CONFIGURATION, CALIBRATION & CHANGE CONTROL

Threshold values and calibration coefficients are controlled configuration data with version, range and integrity checks. A field update is accepted only after compatibility and authenticity checks and cannot alter intrinsic-safety limits or bypass the protected alarm-energy reserve. Rollback and corruption are handled as configuration faults with retained version history.

Failure analysis addresses corrupt or unauthorized configuration creating excessive radio activity or masking alarms. Diagnostic outputs must distinguish bad input, internal malfunction and incomplete engineering evidence, while range-check test; configuration-authentication test; rollback verification provides the objective basis for closing the mechanism.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-CFG-001	Threshold and calibration data shall be versioned, range checked and authenticated according to the released configuration-control method.	Range-check test.
EHSN-CFG-002	Corrupt or unauthorized configuration shall not increase radio activity beyond the energy budget, mask an alarm threshold or alter intrinsic-safety limits.	Misuse-case test for threshold and calibration data ownership plus diagnostic and recovery verification.
EHSN-CFG-003	Threshold or calibration changes shall require range, authenticity and rollback validation before production equivalence.	threshold and calibration data ownership traceability audit with regression-closure review.

Verification, margin and future-change rule

For threshold and calibration data ownership, reassess the release argument when any of these change: threshold package, calibration source, service tool, NVM format or update policy. Regression for threshold and calibration data ownership shall target only the assumptions changed and shall preserve traceability to the prior range-check test evidence rather than repeating unrelated qualification by default.

EVENT DATA CAPTURE, TIMESTAMPING & LOCAL RECORD

Across startup, normal service, degraded operation and authorized recovery, event data capture and minimal local history must preserve the behavior described here. The node stores only the context required to explain alarms, communication attempts and energy state. Data retention is balanced against NVM energy and endurance rather than treated as unlimited logging. State transitions for event data capture and minimal local history are verified from observable conditions rather than inferred from nominal success alone.

Event-history assurance shall demonstrate that logging cannot consume an uncontrolled share of the energy budget and that retained records preserve enough context to interpret an alarm. Verification uses event replay, storage-energy measurement and power-loss integrity testing.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-DAT-001	The architecture shall provide observability for event data capture and minimal local history sufficient to distinguish valid operation from degraded or compromised conditions.	Event replay.
EHSN-DAT-002	When excessive logging consuming energy or missing event context making alarms unactionable occurs, containment shall follow the defined policy and retain the evidence needed for diagnosis.	Fault stimulation of event data capture and minimal local history with interface-state and audit-record review.
EHSN-DAT-003	Acceptance shall be based on event replay; storage-energy measurement; power-loss integrity test, including configuration traceability where it affects the conclusion.	event data capture and minimal local history configuration baseline and qualification-status review.

Verification, margin and future-change rule

For event data capture and minimal local history, reassess the release argument when any of these change: record schema, timestamp source, NVM technology, retention count or diagnostic export. Regression for event data capture and minimal local history shall target only the assumptions changed and shall preserve traceability to the prior event replay evidence rather than repeating unrelated qualification by default.

TRIGGER-TO-ALARM TIMING & STALE-DATA HANDLING

Event timing is measured from the defined sensor trigger boundary through qualification, processing and radio handoff. Exact application latency targets remain controlled parameters unless separately released by the program. Timing evidence is tied to the exact sensor, firmware, radio and energy state used during verification.

The release risk is stale sensor data or queued events being transmitted as if they represent current hazardous conditions. Evidence from timing trace; stale-event injection; queue-overflow test is reviewed together with interface state and recovery behavior; an engineering gap blocks the affected variant rather than being disguised as a runtime derating strategy.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-TIM-001	event timing and freshness shall be dimensioned and configured for the declared mission profile without relying on undocumented external behavior.	Timing trace.
EHSN-TIM-002	The consequence of stale sensor data or queued events being transmitted as if they represent current hazardous conditions shall be bounded so that recovery, fail-closed behavior or service action remains deterministic.	Abnormal-condition test for event timing and freshness and degraded-state trace review.
EHSN-TIM-003	Engineering release shall close timing trace; stale-event injection; queue-overflow test; supplier, layout, firmware or installation changes that alter the argument require focused retest.	Production-equivalence decision record for event timing and freshness.

Verification, margin and future-change rule

For event timing and freshness, reassess the release argument when any of these change: sampling rate, threshold persistence, MCU clock, radio availability or network schedule. Regression for event timing and freshness shall target only the assumptions changed and shall preserve traceability to the prior timing trace evidence rather than repeating unrelated qualification by default.

LORAWAN RADIO VARIANT & ENERGY PROFILE

The product treats the lorawan communication option as a controlled capability rather than a generic platform feature. A LoRaWAN variant may be used for secure low-duty-cycle telemetry. Data rate, transmit power, join behavior and retry policy are controlled because they materially affect energy availability. Any dependency of the lorawan communication option is named in the interface or configuration baseline so its loss can be distinguished from an internal defect.

Radio-energy verification addresses link adaptation, retries and repeated joins that can exhaust the alarm reserve under poor coverage. RF link testing, energy-per-message profiling and outage/rejoin testing quantify this behavior.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-LRA-001	The released implementation of the LoRaWAN communication option shall expose the status required by diagnostics and lifecycle management without disclosing protected secrets or unsafe controls.	Rf link test.
EHSN-LRA-002	Radio adaptation, repeated joins or retry storms that threaten the protected alarm reserve shall be detected by the energy policy and shall curtail nonessential traffic.	Stress/fault test for the lorawan communication option with recovery and non-bypass evidence.
EHSN-LRA-003	Objective evidence for this requirement shall include RF link test; energy-per-message profiling; outage/rejoin test, using pass/fail criteria owned by the product verification plan.	DVP closure for the lorawan communication option and controlled release-board review.

Verification, margin and future-change rule

For the lorawan communication option, reassess the release argument when any of these change: region profile, data rate, TX power, join mode, antenna or gateway coverage. Regression for the lorawan communication option shall target only the assumptions changed and shall preserve traceability to the prior RF link test evidence rather than repeating unrelated qualification by default.

WIRELESSHART RADIO VARIANT & SCHEDULE INTERACTION

A WirelessHART variant may instead use scheduled industrial mesh communication. Slot participation and network maintenance traffic are included in the node energy budget. The engineering objective for the wirelesshart communication option is explicit ownership: integration teams must be able to identify the trusted inputs, permitted outputs and conditions that invalidate the result.

The dominant concern is network schedule or routing churn consuming more energy than the harvester can replenish. The verification package for this page therefore uses network emulator; slot-energy profiling; parent-loss/rejoin test and records the tested revision/configuration so the result can be traced when the design changes.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-WH-001	the WirelessHART communication option shall be implemented with explicit ownership, validity rules and bounded behavior for the released configuration.	Network emulator.
EHSN-WH-002	The design shall detect or contain network schedule or routing churn consuming more energy than the harvester can replenish before it can violate the stated trust, availability or interface boundary.	Targeted the wirelesshart communication option fault injection and diagnostic-state correlation.
EHSN-WH-003	Release of this function shall require network emulator; slot-energy profiling; parent-loss/rejoin test; an unresolved result keeps the affected configuration open.	the WirelessHART communication option release checklist and configuration-status audit.

Verification, margin and future-change rule

For the wirelesshart communication option, reassess the release argument when any of these change: network manager, slot allocation, routing role, channel map, antenna or firmware stack. Regression for the wirelesshart communication option shall target only the assumptions changed and shall preserve traceability to the prior network emulator evidence rather than repeating unrelated qualification by default.

RADIO BURST ENERGY, RETRIES & LINK-MARGIN GOVERNANCE

radio energy consumption and retry governance is a product-owned mechanism with installation dependencies documented separately. The radio is normally the highest instantaneous load. Transmission attempts are therefore admitted by available-energy policy, and retry behavior is bounded so communication failure does not collapse the sensing function. For radio energy consumption and retry governance, that separation keeps deployment choices visible without turning them into hidden assumptions inside the product claim.

Risk analysis centers on poor RF conditions causing uncontrolled retries and complete energy depletion. Instead of inventing a missing margin, the program keeps the unresolved parameter visible and closes it through conducted/RF attenuation test; retry-energy measurement; low-reserve test; operational containment and product-release approval remain separate decisions.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-RFE-001	The product shall preserve the intended function of radio energy consumption and retry governance across the operational states defined for the installation.	Conducted/rf attenuation test.
EHSN-RFE-002	A fault involving poor RF conditions causing uncontrolled retries and complete energy depletion shall produce a diagnosable state and shall not silently expand authority beyond the approved boundary.	Negative-path test for radio energy consumption and retry governance with recovery-state observation.
EHSN-RFE-003	Qualification shall close conducted/RF attenuation test; retry-energy measurement; low-reserve test with results traceable to the tested hardware, software and configuration.	Variant evidence review for radio energy consumption and retry governance and production-approval gate.

Verification, margin and future-change rule

For radio energy consumption and retry governance, reassess the release argument when any of these change: radio module, PA setting, antenna, protocol retry limit, installation loss or network policy. Regression for radio energy consumption and retry governance shall target only the assumptions changed and shall preserve traceability to the prior conducted/RF attenuation test evidence rather than repeating unrelated qualification by default.

AES-128 PAYLOAD PROTECTION & SECURITY SCOPE

AES-128 payload protection is part of the LoRaWAN or WirelessHART product variants. Encryption is used together with the selected network's authentication, freshness, frame-counter and key-lifecycle mechanisms rather than being treated as a standalone security guarantee. Key identifiers and security state are configuration controlled so duplicate, stale or mismatched credentials are detectable during provisioning and field service.

Failure analysis addresses encrypted but replayable, unauthenticated or incorrectly keyed telemetry being accepted as current data. Diagnostic outputs must distinguish bad input, internal malfunction and incomplete engineering evidence, while protocol security test; replay/duplicate test; key-mismatch test provides the objective basis for closing the mechanism.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-AES-001	AES-128 network security shall be integrated with the selected LoRaWAN or WirelessHART authentication, freshness, frame-counter and key-lifecycle mechanisms.	Protocol security test.
EHSN-AES-002	Replayable, unauthenticated or incorrectly keyed telemetry shall not be accepted as current valid data solely because its payload is encrypted.	Boundary-condition campaign for AES-128 payload protection with event-log review.
EHSN-AES-003	Radio-stack or key-management changes shall require protocol security, replay/duplicate and key-mismatch regression before production equivalence.	AES-128 payload protection traceability audit with regression-closure review.

Verification, margin and future-change rule

For AES-128 payload protection, reassess the release argument when any of these change: network stack, key derivation, frame counter, join procedure or backend. Regression for AES-128 payload protection shall target only the assumptions changed and shall preserve traceability to the prior protocol security test evidence rather than repeating unrelated qualification by default.

KEY PROVISIONING, STORAGE & FIELD RENEWAL

Across startup, normal service, degraded operation and authorized recovery, network key lifecycle must preserve the behavior described here. Network credentials are provisioned and stored so routine service does not expose reusable secrets. Renewal or replacement follows the selected LoRaWAN/WirelessHART trust model and preserves device identity. State transitions for network key lifecycle are verified from observable conditions rather than inferred from nominal success alone.

Network-key assurance shall demonstrate that shared, duplicated or stale keys cannot provide unintended network admission or device impersonation. Verification uses provisioning audit, duplicate-key checks, replay testing and renewal/rejoin validation.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-KEY-001	The architecture shall provide observability for network key lifecycle sufficient to distinguish valid operation from degraded or compromised conditions.	Provisioning audit.
EHSN-KEY-002	When shared, duplicated or stale keys allowing impersonation or loss of network admission occurs, containment shall follow the defined policy and retain the evidence needed for diagnosis.	Injected-failure test for network key lifecycle with containment and restart evidence.
EHSN-KEY-003	Acceptance shall be based on provisioning audit; duplicate-key check; renewal/rejoin test, including configuration traceability where it affects the conclusion.	network key lifecycle configuration baseline and qualification-status review.

Verification, margin and future-change rule

For network key lifecycle, reassess the release argument when any of these change: key source, manufacturing station, secure storage, network manager or service workflow. Regression for network key lifecycle shall target only the assumptions changed and shall preserve traceability to the prior provisioning audit evidence rather than repeating unrelated qualification by default.

FAILURE MODES, DEGRADED OPERATION & ALARM PRIORITY

Engineering ownership of degraded operation under low energy or sensor/network faults includes its dependencies and failure evidence. When energy is insufficient for full telemetry, the node prioritizes event sensing and the reserved alarm path according to the approved mission. It reports degraded energy or communication state when connectivity permits. The release record therefore ties degraded operation under low energy or sensor/network faults to the exact implementation and configured external services used in verification.

The release risk is nonessential activity continuing during scarcity and preventing the emergency alarm attempt. Evidence from low-energy scenario test; sensor/radio fault injection; priority-state review is reviewed together with interface state and recovery behavior; an engineering gap blocks the affected variant rather than being disguised as a runtime derating strategy.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-FAL-001	degraded operation under low energy or sensor/network faults shall be dimensioned and configured for the declared mission profile without relying on undocumented external behavior.	Low-energy scenario test.
EHSN-FAL-002	The consequence of nonessential activity continuing during scarcity and preventing the emergency alarm attempt shall be bounded so that recovery, fail-closed behavior or service action remains deterministic.	Misuse-case test for degraded operation under low energy or sensor/network faults plus diagnostic and recovery verification.
EHSN-FAL-003	Engineering release shall close low-energy scenario test; sensor/radio fault injection; priority-state review; supplier, layout, firmware or installation changes that alter the argument require focused retest.	Production-equivalence decision record for degraded operation under low energy or sensor/network faults.

Verification, margin and future-change rule

For degraded operation under low energy or sensor/network faults, reassess the release argument when any of these change: energy policy, task priority, network retry policy, sensor fault handling or diagnostics. Regression for degraded operation under low energy or sensor/network faults shall target only the assumptions changed and shall preserve traceability to the prior low-energy scenario test evidence rather than repeating unrelated qualification by default.

REDUNDANT STORAGE SINGLE-FAULT CONTAINMENT

The product treats single-fault behavior of redundant storage as a controlled capability rather than a generic platform feature. Redundant capacitors are credited only when open, short and high-leakage faults are considered. Isolation and monitoring ensure a failed element cannot silently drain the companion reserve. Any dependency of single-fault behavior of redundant storage is named in the interface or configuration baseline so its loss can be distinguished from an internal defect.

Storage-redundancy verification addresses common charging paths and leakage coupling that could defeat nominal redundancy. Single-fault circuit testing, leakage injection and reserve-energy measurement demonstrate isolation between storage elements.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-RED-001	The released implementation of single-fault behavior of redundant storage shall expose the status required by diagnostics and lifecycle management without disclosing protected secrets or unsafe controls.	Single-fault circuit test.
EHSN-RED-002	Common charging paths or leakage coupling shall not defeat storage redundancy or allow one failed storage element to drain the protected reserve.	Fault stimulation of single-fault behavior of redundant storage with interface-state and audit-record review.
EHSN-RED-003	Objective evidence for this requirement shall include single-fault circuit test; leakage injection; reserve-energy verification, using pass/fail criteria owned by the product verification plan.	DVP closure for single-fault behavior of redundant storage and controlled release-board review.

Verification, margin and future-change rule

For single-fault behavior of redundant storage, reassess the release argument when any of these change: storage topology, isolation resistor/FET, charger, monitor circuit or PCB contamination. Regression for single-fault behavior of redundant storage shall target only the assumptions changed and shall preserve traceability to the prior single-fault circuit test evidence rather than repeating unrelated qualification by default.

-40 C TO +85 C POWER, STORAGE & SENSOR OPERATION

The released ambient operating envelope is -40 C to +85 C. Harvesting efficiency, storage ESR/leakage, MCU startup, sensor accuracy and radio energy shall be evaluated across that range. The environmental qualification matrix identifies the controlled inputs, permitted outputs, conditioning states and post-stress acceptance criteria for each released variant.

The dominant concern is cold ESR, hot leakage or harvester efficiency shifts invalidating the room-temperature energy budget. The verification package for this page therefore uses temperature-chamber energy test; cold-start test; hot-leakage characterization and records the tested revision/configuration so the result can be traced when the design changes.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-TMP-001	the stated -40 C to +85 C ambient envelope shall be implemented with explicit ownership, validity rules and bounded behavior for the released configuration.	Temperature-chamber energy test.
EHSN-TMP-002	The design shall detect or contain cold ESR, hot leakage or harvester efficiency shifts invalidating the room-temperature energy budget before it can violate the stated trust, availability or interface boundary.	Abnormal-condition test for the stated -40 C to +85 C ambient envelope and degraded-state trace review.
EHSN-TMP-003	Release of this function shall require temperature-chamber energy test; cold-start test; hot-leakage characterization; an unresolved result keeps the affected configuration open.	the stated -40 C to +85 C ambient envelope release checklist and configuration-status audit.

Verification, margin and future-change rule

For the stated -40 C to +85 C ambient envelope, reassess the release argument when any of these change: storage cell, harvester, PMIC, MCU, sensor, radio, enclosure or installation. Regression for the stated -40 C to +85 C ambient envelope shall target only the assumptions changed and shall preserve traceability to the prior temperature-chamber energy test evidence rather than repeating unrelated qualification by default.

INTRINSIC-SAFETY ENERGY LIMITATION & FAULT ANALYSIS

intrinsic-safety energy limitation is a product-owned mechanism with installation dependencies documented separately. The node is intended for an Ex i enclosure in Zone 1 hazardous areas. The released design must bound available electrical and stored energy under the applicable fault assumptions defined by the certification program. For intrinsic-safety energy limitation, that separation keeps deployment choices visible without turning them into hidden assumptions inside the product claim.

Risk analysis centers on stored or fault energy exceeding the approved intrinsic-safety protection concept. Instead of inventing a missing margin, the program keeps the unresolved parameter visible and closes it through intrinsic-safety analysis; component-fault test; controlled certification evidence; operational containment and product-release approval remain separate decisions.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-EXI-001	The product shall preserve the intended function of intrinsic-safety energy limitation across the operational states defined for the installation.	Intrinsic-safety analysis.
EHSN-EXI-002	A fault involving stored or fault energy exceeding the approved intrinsic-safety protection concept shall produce a diagnosable state and shall not silently expand authority beyond the approved boundary.	Stress/fault test for intrinsic-safety energy limitation with recovery and non-bypass evidence.
EHSN-EXI-003	Qualification shall close intrinsic-safety analysis; component-fault test; controlled certification evidence with results traceable to the tested hardware, software and configuration.	Variant evidence review for intrinsic-safety energy limitation and production-approval gate.

Verification, margin and future-change rule

For intrinsic-safety energy limitation, reassess the release argument when any of these change: storage value, voltage limit, current-limiting components, spacing, encapsulation or certification basis. Regression for intrinsic-safety energy limitation shall target only the assumptions changed and shall preserve traceability to the prior intrinsic-safety analysis evidence rather than repeating unrelated qualification by default.

Zone 1 ENCLOSURE & HAZARDOUS-AREA INSTALLATION

The Zone 1 mechanical installation includes the enclosure, mounting, cable/antenna interfaces and any accessories that form part of the intrinsic-safety concept. Feedthrough, antenna or accessory substitutions require hazardous-area impact review because they can change stored energy, fault paths or RF behavior. Installation instructions identify the approved combinations and their configuration evidence.

Failure analysis addresses field wiring, antenna or enclosure modification invalidating the hazardous-area protection concept. Diagnostic outputs must distinguish bad input, internal malfunction and incomplete engineering evidence, while installation review; enclosure test; accessory/configuration audit provides the objective basis for closing the mechanism.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-Z1-001	Zone 1 mechanical and installation controls shall identify the approved enclosure, antenna/feedthrough, cabling and accessories that are part of the intrinsic-safety concept.	Installation review.
EHSN-Z1-002	Field wiring, antenna or enclosure modifications that affect the hazardous-area protection concept shall require impact review and requalification before the modified installation is approved.	Targeted Zone 1 mechanical and installation controls fault injection and diagnostic-state correlation.
EHSN-Z1-003	Enclosure, antenna or installation-accessory changes shall require hazardous-area installation review and configuration qualification before release.	Zone 1 mechanical and installation controls traceability audit with regression-closure review.

Verification, margin and future-change rule

For Zone 1 mechanical and installation controls, reassess the release argument when any of these change: enclosure, gland, antenna, mounting, sensor lead, field connector or installation instruction. Regression for Zone 1 mechanical and installation controls shall target only the assumptions changed and shall preserve traceability to the prior installation review evidence rather than repeating unrelated qualification by default.

HAZARDOUS-AREA SENSOR / HARVESTER INTERFACE BOUNDARY

Across startup, normal service, degraded operation and authorized recovery, hazardous-area interface energy and fault containment must preserve the behavior described here. Energy harvesters and external sensing interfaces cross mechanical boundaries that can conduct energy into the enclosure. Their maximum credible electrical contribution is included in the intrinsic-safety assessment. State transitions for hazardous-area interface energy and fault containment are verified from observable conditions rather than inferred from nominal success alone.

Intrinsic-safety interface analysis shall demonstrate that external harvester or sensor faults cannot inject more energy than the protected internal circuit assumes. Verification uses interface fault analysis, source-limit testing and installation-envelope review.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-HAZ-001	The architecture shall provide observability for hazardous-area interface energy and fault containment sufficient to distinguish valid operation from degraded or compromised conditions.	Interface fault analysis.
EHSN-HAZ-002	When external harvester or sensor fault injecting more energy than the protected internal circuit assumes occurs, containment shall follow the defined policy and retain the evidence needed for diagnosis.	Negative-path test for hazardous-area interface energy and fault containment with recovery-state observation.
EHSN-HAZ-003	Acceptance shall be based on interface fault analysis; source-limit test; installation-envelope review, including configuration traceability where it affects the conclusion.	hazardous-area interface energy and fault containment configuration baseline and qualification-status review.

Verification, margin and future-change rule

For hazardous-area interface energy and fault containment, reassess the release argument when any of these change: harvester type, cable, sensor, grounding, connector, external capacitance/inductance or mounting. Regression for hazardous-area interface energy and fault containment shall target only the assumptions changed and shall preserve traceability to the prior interface fault analysis evidence rather than repeating unrelated qualification by default.

ENCLOSURE, PIPE MOUNTING, SHOCK, VIBRATION & CHEMICAL EXPOSURE

Engineering ownership of mechanical robustness for remote pipe installation includes its dependencies and failure evidence. Mechanical design must hold the harvester coupling and antenna geometry while tolerating industrial vibration, handling and environmental contamination. Exact qualification severities remain program controlled unless supplied. The release record therefore ties mechanical robustness for remote pipe installation to the exact implementation and configured external services used in verification.

The release risk is mount loosening changing both energy capture and sensing response, or chemicals degrading seals and potting. Evidence from mechanical endurance; chemical exposure; post-test energy/sensor correlation is reviewed together with interface state and recovery behavior; an engineering gap blocks the affected variant rather than being disguised as a runtime derating strategy.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-MEC-001	mechanical robustness for remote pipe installation shall be dimensioned and configured for the declared mission profile without relying on undocumented external behavior.	Mechanical endurance.
EHSN-MEC-002	The consequence of mount loosening changing both energy capture and sensing response, or chemicals degrading seals and potting shall be bounded so that recovery, fail-closed behavior or service action remains deterministic.	Boundary-condition campaign for mechanical robustness for remote pipe installation with event-log review.
EHSN-MEC-003	Engineering release shall close mechanical endurance; chemical exposure; post-test energy/sensor correlation; supplier, layout, firmware or installation changes that alter the argument require focused retest.	Production-equivalence decision record for mechanical robustness for remote pipe installation.

Verification, margin and future-change rule

For mechanical robustness for remote pipe installation, reassess the release argument when any of these change: mount, enclosure material, fastener, adhesive, seal, coating or pipe interface. Regression for mechanical robustness for remote pipe installation shall target only the assumptions changed and shall preserve traceability to the prior mechanical endurance evidence rather than repeating unrelated qualification by default.

ANTENNA, RF FEEDTHROUGH & INTRINSIC-SAFETY INTERACTION

The product treats antenna/rf integration within the hazardous-area concept as a controlled capability rather than a generic platform feature. The radio antenna and feedthrough are part of both RF performance and hazardous-area configuration. Changes can alter link margin, retry energy and the certified installation boundary. Any dependency of antenna/rf integration within the hazardous-area concept is named in the interface or configuration baseline so its loss can be distinguished from an internal defect.

Antenna and installation verification addresses substitutions that increase retries, change RF range or invalidate the approved field-energy model. RF chamber/site testing, installation inspection and configuration audit establish the released antenna variant.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-ANT-001	The released implementation of antenna/RF integration within the hazardous-area concept shall expose the status required by diagnostics and lifecycle management without disclosing protected secrets or unsafe controls.	Rf chamber/site test.
EHSN-ANT-002	Antenna substitution that increases retry energy or changes RF performance shall require updated radio-energy and installation evidence before production or field approval.	Injected-failure test for antenna/rf integration within the hazardous-area concept with containment and restart evidence.
EHSN-ANT-003	Objective evidence for this requirement shall include RF chamber/site test; installation inspection; configuration audit, using pass/fail criteria owned by the product verification plan.	DVP closure for antenna/rf integration within the hazardous-area concept and controlled release-board review.

Verification, margin and future-change rule

For antenna/rf integration within the hazardous-area concept, reassess the release argument when any of these change: antenna type, cable, connector, matching network, enclosure placement or certification variant. Regression for antenna/rf integration within the hazardous-area concept shall target only the assumptions changed and shall preserve traceability to the prior RF chamber/site test evidence rather than repeating unrelated qualification by default.

EMC / RF COEXISTENCE & FALSE-TRIGGER RESILIENCE

Industrial radios and nearby equipment can couple into high-impedance sensing and energy-harvesting nodes. Immunity testing therefore checks false triggers, MCU resets and energy-accounting corruption as well as basic communication continuity. The engineering objective for electromagnetic compatibility and coexistence is explicit ownership: integration teams must be able to identify the trusted inputs, permitted outputs and conditions that invalidate the result.

The dominant concern is RF or conducted interference creating false alarms, missed events or repeated energy-expensive wakeups. The verification package for this page therefore uses EMC immunity; coexistence test; event-rate/energy monitoring and records the tested revision/configuration so the result can be traced when the design changes.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-EMC-001	electromagnetic compatibility and coexistence shall be implemented with explicit ownership, validity rules and bounded behavior for the released configuration.	Emc immunity.
EHSN-EMC-002	The design shall detect or contain RF or conducted interference creating false alarms, missed events or repeated energy-expensive wakeups before it can violate the stated trust, availability or interface boundary.	Misuse-case test for electromagnetic compatibility and coexistence plus diagnostic and recovery verification.
EHSN-EMC-003	Release of this function shall require EMC immunity; coexistence test; event-rate/energy monitoring; an unresolved result keeps the affected configuration open.	electromagnetic compatibility and coexistence release checklist and configuration-status audit.

Verification, margin and future-change rule

For electromagnetic compatibility and coexistence, reassess the release argument when any of these change: PCB layout, filter, antenna, sensor cable, firmware debounce or installation grounding. Regression for electromagnetic compatibility and coexistence shall target only the assumptions changed and shall preserve traceability to the prior EMC immunity evidence rather than repeating unrelated qualification by default.

EOL HARVESTER EFFICIENCY & ENERGY-PATH VALIDATION

end-of-line validation of energy-harvesting efficiency is a product-owned mechanism with installation dependencies documented separately. Manufacturing EOL verifies that the transducer and conditioning path produce the expected response under a controlled stimulus. The measurement is normalized to the applied vibration or thermal gradient. For end-of-line validation of energy-harvesting efficiency, that separation keeps deployment choices visible without turning them into hidden assumptions inside the product claim.

Risk analysis centers on weak transducer coupling, wrong component population or excessive leakage escaping assembly test. Instead of inventing a missing margin, the program keeps the unresolved parameter visible and closes it through calibrated stimulus test; station MSA; golden-unit correlation; operational containment and product-release approval remain separate decisions.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-EOL-001	The product shall preserve the intended function of end-of-line validation of energy-harvesting efficiency across the operational states defined for the installation.	Calibrated stimulus test.
EHSN-EOL-002	A fault involving weak transducer coupling, wrong component population or excessive leakage escaping assembly test shall produce a diagnosable state and shall not silently expand authority beyond the approved boundary.	Fault stimulation of end-of-line validation of energy-harvesting efficiency with interface-state and audit-record review.
EHSN-EOL-003	Qualification shall close calibrated stimulus test; station MSA; golden-unit correlation with results traceable to the tested hardware, software and configuration.	Variant evidence review for end-of-line validation of energy-harvesting efficiency and production-approval gate.

Verification, margin and future-change rule

For end-of-line validation of energy-harvesting efficiency, reassess the release argument when any of these change: EOL fixture, stimulus, transducer lot, PMIC, storage, calibration factor or station software. Regression for end-of-line validation of energy-harvesting efficiency shall target only the assumptions changed and shall preserve traceability to the prior calibrated stimulus test evidence rather than repeating unrelated qualification by default.

MANUFACTURING CTQ, CLEANLINESS & LEAKAGE CONTROL

At microwatt scale, flux residue, moisture, contamination and assembly damage can create leakage comparable to the intended sleep current. Cleanliness, board leakage and process contamination are therefore manufacturing CTQs. Production acceptance is based on process capability, controlled cleanliness/leakage measurements and station correlation; manufacturing evidence is a release gate, not a runtime product state.

Process contamination that can create a field energy deficit shall be detected through ionic/cleanliness controls, board-leakage testing and process-capability monitoring. A unit or process lot that does not satisfy the released acceptance criteria is contained from production release; runtime diagnostics are not used as a substitute for manufacturing conformity.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-MFG-001	Manufacturing shall control cleanliness, board leakage and other CTQs that materially affect the microwatt energy budget.	Ionic/cleanliness control, leakage measurement and station MSA.
EHSN-MFG-002	Units or process lots that fail released leakage or contamination acceptance criteria shall be contained from production release.	Board-leakage test, process capability review and containment record.
EHSN-MFG-003	Manufacturing-process or supplier changes shall require leakage/cleanliness correlation and process-capability closure before production equivalence.	Process-change review with correlation and capability closure.

Verification, margin and future-change rule

For manufacturing controls for ultra-low-power performance, reassess the release argument when any of these change: cleaning process, coating, PCB finish, component handling, humidity conditioning or supplier. Regression for manufacturing controls for ultra-low-power performance shall target only the assumptions changed and shall preserve traceability to the prior ionic/cleanliness control evidence rather than repeating unrelated qualification by default.

SUPERCAPACITOR AGING & THOUSANDS-OF-CYCLES MODEL

Across startup, normal service, degraded operation and authorized recovery, storage aging over thousands of charge cycles must preserve the behavior described here. The lifecycle model includes thousands of charge cycles rather than assuming a new storage element. Capacity fade, ESR growth and leakage increase are correlated to temperature and actual cycling profile. State transitions for storage aging over thousands of charge cycles are verified from observable conditions rather than inferred from nominal success alone.

Lifecycle validation shall demonstrate that aged storage can still deliver the required alarm pulse energy when the open-circuit voltage alone appears acceptable. Verification uses accelerated cycling, pulse-load testing and model-to-test correlation.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-AGE-001	The architecture shall provide observability for storage aging over thousands of charge cycles sufficient to distinguish valid operation from degraded or compromised conditions.	Accelerated cycling.
EHSN-AGE-002	When aged storage failing to deliver alarm pulse energy even though open-circuit voltage appears acceptable occurs, containment shall follow the defined policy and retain the evidence needed for diagnosis.	Stress/fault test for storage aging over thousands of charge cycles with recovery and non-bypass evidence.
EHSN-AGE-003	Acceptance shall be based on accelerated cycling; pulse-load test; model-to-test correlation, including configuration traceability where it affects the conclusion.	storage aging over thousands of charge cycles configuration baseline and qualification-status review.

Verification, margin and future-change rule

For storage aging over thousands of charge cycles, reassess the release argument when any of these change: storage chemistry, charge window, temperature, cycle depth, leakage or mission profile. Regression for storage aging over thousands of charge cycles shall target only the assumptions changed and shall preserve traceability to the prior accelerated cycling evidence rather than repeating unrelated qualification by default.

15-YEAR MAINTENANCE-FREE LIFECYCLE & ENERGY AVAILABILITY

Engineering ownership of the 15-year maintenance-free mission includes its dependencies and failure evidence. The long-life claim depends on energy-harvester stability, storage aging, sensor drift, radio-network evolution and mechanical coupling. Field diagnostics identify degradation trends before monitoring becomes unreliable. The release record therefore ties the 15-year maintenance-free mission to the exact implementation and configured external services used in verification.

The release risk is lifecycle drift converting a maintenance-free concept into silent loss of detection or communication. Evidence from lifecycle model; accelerated aging; field-health metric review is reviewed together with interface state and recovery behavior; an engineering gap blocks the affected variant rather than being disguised as a runtime derating strategy.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-LIF-001	the 15-year maintenance-free mission shall be dimensioned and configured for the declared mission profile without relying on undocumented external behavior.	Lifecycle model.
EHSN-LIF-002	The consequence of lifecycle drift converting a maintenance-free concept into silent loss of detection or communication shall be bounded so that recovery, fail-closed behavior or service action remains deterministic.	Targeted the 15-year maintenance-free mission fault injection and diagnostic-state correlation.
EHSN-LIF-003	Engineering release shall close lifecycle model; accelerated aging; field-health metric review; supplier, layout, firmware or installation changes that alter the argument require focused retest.	Production-equivalence decision record for the 15-year maintenance-free mission.

Verification, margin and future-change rule

For the 15-year maintenance-free mission, reassess the release argument when any of these change: harvester, storage, sensor, antenna, firmware, network or mounting interface. Regression for the 15-year maintenance-free mission shall target only the assumptions changed and shall preserve traceability to the prior lifecycle model evidence rather than repeating unrelated qualification by default.

ENERGY-SCENARIO DVP & END-TO-END ALARM VALIDATION

The integrated DVP is a release-evidence set spanning energy harvesting, sensing, storage, radio behavior and fault cases. It replays representative weak/strong energy sources, temperature, storage aging, sensor events and radio coverage using the exact product configuration intended for release. Alarm success is judged from the defined trigger boundary, measured energy reserve and network state.

End-to-end DVP addresses the case where subsystems pass independently but the assembled node cannot complete sensing and alarm transmission under realistic energy scarcity. Scenario-matrix testing and fault injection shall exercise the released harvester, storage age, radio network and firmware together.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-DVP-001	The released implementation of the integrated DVP across energy, sensing, radio and faults shall expose the status required by diagnostics and lifecycle management without disclosing protected secrets or unsafe controls.	Scenario matrix.
EHSN-DVP-002	The integrated product shall not be released when subsystem evidence passes individually but end-to-end sensing and alarm transmission cannot be demonstrated under the representative energy-scarcity scenarios.	Negative-path test for the integrated DVP across energy, sensing, radio and faults with recovery-state observation.
EHSN-DVP-003	Objective evidence for this requirement shall include scenario matrix; end-to-end alarm tests; fault-injection coverage audit, using pass/fail criteria owned by the product verification plan.	DVP closure for the integrated DVP across energy, sensing, radio and faults and controlled release-board review.

Verification, margin and future-change rule

For the integrated DVP across energy, sensing, radio and faults, reassess the release argument when any of these change: mission profile, harvester replay, storage age, radio network, firmware or sensor configuration. Regression for the integrated DVP across energy, sensing, radio and faults shall target only the assumptions changed and shall preserve traceability to the prior scenario matrix evidence rather than repeating unrelated qualification by default.

ENVIRONMENTAL / HAZARDOUS-AREA QUALIFICATION RELEASE GATE

Environmental and intrinsic-safety qualification are engineering release gates. Missing evidence does not cause a fictitious runtime derate; the affected hardware or installation variant remains unreleased. The engineering objective for environmental and hazardous-area qualification evidence is explicit ownership: integration teams must be able to identify the trusted inputs, permitted outputs and conditions that invalidate the result.

The dominant concern is an unqualified enclosure, component or installation entering production based only on similarity assumptions. The verification package for this page therefore uses qualification matrix; certification evidence review; configuration-status audit and records the tested revision/configuration so the result can be traced when the design changes.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-ENV-001	environmental and hazardous-area qualification evidence shall be implemented with explicit ownership, validity rules and bounded behavior for the released configuration.	Qualification matrix.
EHSN-ENV-002	The design shall detect or contain an unqualified enclosure, component or installation entering production based only on similarity assumptions before it can violate the stated trust, availability or interface boundary.	Boundary-condition campaign for environmental and hazardous-area qualification evidence with event-log review.
EHSN-ENV-003	Release of this function shall require qualification matrix; certification evidence review; configuration-status audit; an unresolved result keeps the affected configuration open.	environmental and hazardous-area qualification evidence release checklist and configuration-status audit.

Verification, margin and future-change rule

For environmental and hazardous-area qualification evidence, reassess the release argument when any of these change: enclosure, storage, harvester, antenna, sensor, PCB material, installation or certification scope. Regression for environmental and hazardous-area qualification evidence shall target only the assumptions changed and shall preserve traceability to the prior qualification matrix evidence rather than repeating unrelated qualification by default.

SUPPLIER, SECOND-SOURCE & PRODUCTION-EQUIVALENCE CLOSURE

Alternate harvesters, storage capacitors, radios, sensors or enclosure materials are not production-equivalent by declaration. Each substitution is assessed for harvested energy, leakage, intrinsic-safety fault energy, RF behavior and environmental impact, and the affected qualification/regression evidence is closed before production approval.

Second-source risk centers on a substitution reducing harvested energy, increasing leakage or stored/fault energy, or changing radio/environmental behavior. Delta analysis, targeted regression and release-board approval determine equivalence; unresolved evidence blocks the affected variant rather than being represented as a runtime condition.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-SRC-001	The product shall preserve the intended function of supplier and second-source equivalence across the operational states defined for the installation.	Delta analysis.
EHSN-SRC-002	A fault involving second-source change reducing harvested energy or increasing stored/fault energy without targeted validation shall produce a diagnosable state and shall not silently expand authority beyond the approved boundary.	Injected-failure test for supplier and second-source equivalence with containment and restart evidence.
EHSN-SRC-003	Qualification shall close delta analysis; targeted regression; release-board approval with results traceable to the tested hardware, software and configuration.	Variant evidence review for supplier and second-source equivalence and production-approval gate.

Verification, margin and future-change rule

For supplier and second-source equivalence, reassess the release argument when any of these change: harvester, storage, PMIC, radio, sensor, enclosure, PCB or manufacturing supplier. Regression for supplier and second-source equivalence shall target only the assumptions changed and shall preserve traceability to the prior delta analysis evidence rather than repeating unrelated qualification by default.

FINAL RELEASE CLOSURE & FUTURE-CHANGE GOVERNANCE

Final release closure and future-change governance are engineering controls. Release links the energy budget, hazardous-area evidence, RF/security behavior, EOL calibration and lifecycle assumptions to the exact hardware/software/installation variant. Open parameters remain owned and visible, and any change affecting the energy or intrinsic-safety argument blocks production equivalence until the required impact analysis and regression are closed.

Failure analysis addresses future updates or substitutions invalidating the energy or intrinsic-safety argument without triggering requalification. Diagnostic outputs must distinguish bad input, internal malfunction and incomplete engineering evidence, while release checklist; open-item audit; change-impact and regression review provides the objective basis for closing the mechanism.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
EHSN-CLS-001	Final release shall link energy, sensing, radio, intrinsic-safety, EOL and lifecycle evidence to the exact product and installation variant.	Release checklist and configuration-status accounting.
EHSN-CLS-002	A design or supplier change that invalidates the approved energy or intrinsic-safety argument shall block release until impact analysis and required requalification are closed.	Change-impact review, open-item audit and certification/qualification disposition.
EHSN-CLS-003	Any change affecting the approved energy or intrinsic-safety argument shall require impact analysis and targeted release evidence before production equivalence.	Targeted DVP closure and controlled release-board approval.

Verification, margin and future-change rule

For final release closure and change governance, reassess the release argument when any of these change: any change to energy path, storage, sensor, radio, enclosure, intrinsic-safety protection, firmware or mission profile. Regression for final release closure and change governance shall target only the assumptions changed and shall preserve traceability to the prior release checklist evidence rather than repeating unrelated qualification by default.